

SUPPORTING INFORMATION

Spike Growth on Patterned Gold Nanoparticle Scaffolds

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METHODS

Spike growth

EPPS buffer (20 mL) with a designed concentration and pH was added into a 30-mL beaker containing a magnetic stirring bar. The substrate with a region of annealed patterned AuNPs was then vertically suspended in the beaker by hanging on a laboratory support stand using scissor-type forceps, ensuring that the NP region was positioned 1 cm away from the stirring bar. Initially, the EPPS buffer underwent magnetic stirring for 1 h at 800 rpm. Following this, HAuCl₄ (320 μ L) was injected into the solution to reach a final concentration of 0.4 mM, and magnetic stirring continued for another hour. The total volume of the growth solution amounted to 20 mL. After the stirring process, the system was left unattended for the designed growth time points for the growth mechanism experiment (Figure 3) and 16 h for all other experiments. Spike growth can continue up to 90 min; after this time, the particle sizes show little change since the gold precursor (HAuCl₄) in the solution is depleted from the simultaneous formation of colloidal gold nanostars.^{1, 2} Longer incubation times can assist in producing spiky AuNP arrays without undesired nanostars on the substrate since they will settle to the bottom of the reaction beaker under gravity. Finally, the patterned AuNPs were washed thoroughly with deionized water and dried with nitrogen gas.

Scanning electron microscopy

Scanning electron microscope (SEM) images were obtained with a Hitachi S4800-II eFEG SEM. The beam source derived from a cold field emitter with an applied accelerating voltage of 25 kV and an emission current of 5 μ A.

Morphology analysis

The core diameter and spike length of AuNPs were determined manually from the SEM images (Figure 2a-d, S3a-d, 5a-d) using the ImageJ analysis software. For the core diameter, an elliptical

shape was applied to 48 particles per condition, and the average of the largest and smallest diameters of this ellipse was used as the core diameter (**Figure 2e**). For the spike length, the distance between the spike tip to the middle transition of the core was measured. At least 50 particles were measured for each condition for the spike length analysis.

To address the challenge of accurately determining the spike numbers of three-dimensional (3D) nanostructures from two-dimensional (2D) SEM top-view images, we used a normalized peripheral spike number as a comparative metric. Specifically, we focused on counting the number of peripheral spikes outside of the core (green ring in **Figure 2f**). Since the spikes are randomly oriented, the peripheral spike number can represent the overall degree of branching in the entire AuNP. The number of peripheral spikes can also avoid the challenge of counting overlapped spikes from the projected 2D SEM images. For **Figure 2f**, **Figure S4b**, and **Figure S7b**, the peripheral spike numbers were normalized with the highest peripheral spike number (24 spikes/particle) of spiky AuNPs synthesized with initial seeds with diameters of 70 nm. 48 particles were measured per condition.

Dark-field spectroscopy

We performed dark-field microscopy and spectroscopy with an inverted Nikon TE2000-U microscope. This setup featured an oil immersion $\times 100$ objective (NA = 0.7-1.3) and an oil condenser (NA = 1.2-1.5). The light source was an unfiltered and unpolarized 100 W tungsten-halogen lamp. The transmitted light was directed to a spectrometer (Acton SP2300, Princeton Instruments) equipped with a liquid nitrogen-cooled charged coupled device (CCD, Princeton Instruments). The substrate of spiky AuNPs was fixed onto the sample stage and covered with a quartz coverslip. Index oil was applied between the objective and coverslip, and between the condenser and the backside of the substrate. We recorded dark-field scattering spectra at the same

light intensity and aperture, with the exposure time of 500 ms. The dark-field scattering spectra were collected from 5 spots per condition.

Raman spectroscopy

Spiky AuNP arrays were immersed in a 4-mercaptobenzonitrile (1 mM) solution (ethanol as solvent) and incubated for 16 h to form a self-assembled monolayer on the surface and then rinsed with ethanol thoroughly and dried with N₂ gas. The Raman spectra were measured with a 633-nm excitation laser using a confocal Raman microscope (LabRAM HR Evolution, Horiba). The excitation power was kept at 2 mW. The acquisition time was 2 s with the accumulation of 10. Raman spectra were collected with a $\times 100$ objective (NA = 0.8) and dispersed by a diffraction grating (600 lines/mm). The Raman spectra were collected from 5 individual AuNPs per condition and showed similar intensities of Raman signals.

Simulations

The electromagnetic near-field distribution was calculated using finite-difference time domain (FDTD) methods. The model structure was based on the morphology of spiky AuNPs synthesized with the initial seed diameter of 70 nm and EPPS buffer (400 mM) at pH 7.5. The orientation of spikes was randomly distributed to make the model structure similar to spiky AuNPs as imaged in the SEM. The refractive index of gold was from Palik,³ and the refractive index of quartz substrate was set to 1.45. A mesh size of 1 nm was applied to the entire structure. Total-field scattered-field light source was injected from the quartz substrate side with normal incidence angle, and near-field profiles were collected by frequency-domain field and power monitors.

Fabrication of the patterned AuNPs

Periodic photoresist posts were generated by solvent-assisted nanoscale embossing (SANE) (**Figure S1a**).⁴ First, an adhesion layer (1:4 HMDS : PGMEA) and a layer of positive photoresist (1:8 photoresist : PGMEA) were spin-coated on silicon (Si) wafers and baked at 120 °C for 90 s. Then, a PDMS mask (periodicity = 600 nm) was immersed in DMF for 90 s and placed on the photoresist-coated wafer for 1 h. Residual resist between the as-prepared photoresist posts was removed with O₂ reactive ion etching (RIE) (50 sccm O₂, 20 Pa, 70 W, 30 s) (Samco RIE-10NR).

Au hole masks were generated by PEEL (photolithography, etching, electron-beam metal deposition, and lift-off) (**Figure S1b**).⁵ First, 8 nm of Cr was deposited on the periodic photoresist posts using a thermal evaporator (Nano 36, Kurt J. Lesker Company). To fabricate the Si hole array, a 16 h lift-off process was carried out using photoresist remover 1165, followed by a second RIE process (20 sccm SF₆, 40 sccm CF₄, 10 sccm O₂, 3 Pa, 30 W, 8 min). 80 nm of Cr and 140 nm of Au were deposited on the Si hole array to create the Au hole mask. To transfer the Au hole mask to the substrate, the Cr layer was wet etched in Cr etchant for 45 min to release the mask from the Si wafer. The mask was then floated on the water surface and transferred onto the desired substrate. For samples intended for SEM imaging, ITO-quartz substrates were used to prevent charging effects. Quartz substrates were used for samples for optical experiments. Finally, 5 nm of Cr and 90 nm of Au were deposited onto the substrates through the Au holes of the mask. The mask was then removed by Scotch tape to create the patterned AuNPs (**Figure S1c**).

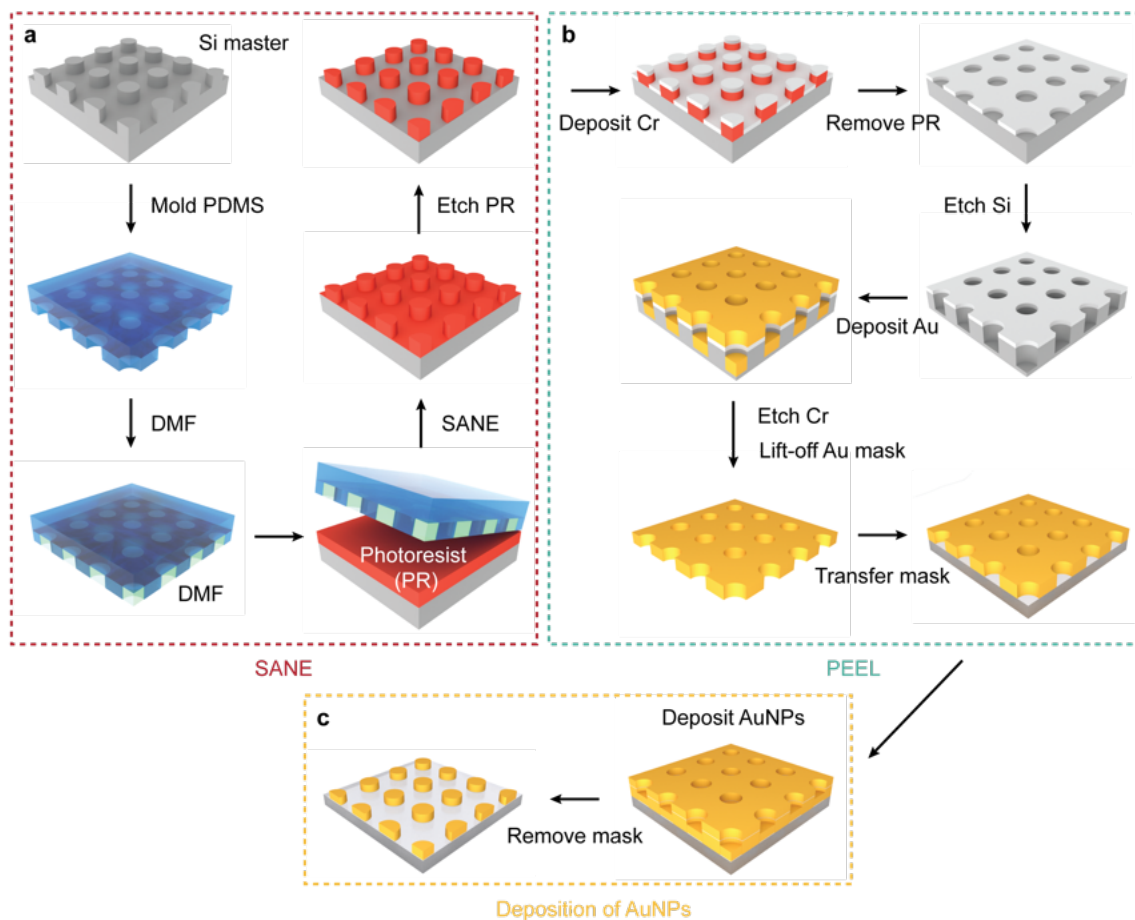


Figure S1. Schematic fabrication procedure of the patterned AuNPs. (a) SANE process showing the procedures to create periodic photoresist posts. **(b)** PEEL process depicting the fabrication procedures of nanohole arrays. **(c)** Deposition process to create the patterned AuNPs.

Thermal annealing of as-fabricated AuNPs

The annealing process was performed in an MTI furnace system (OTF-1200x). As-fabricated patterned AuNPs were placed in a quartz tube. The reaction chamber was evacuated and flushed with 25 sccm of argon (Ar) gas at the total pressure of 1.2 torr for 5 min. The temperature was then increased to 800 °C with a 10 °C/min ramp followed by annealing for 30 min. Subsequently, the temperature of the chamber was cooled down to room temperature.

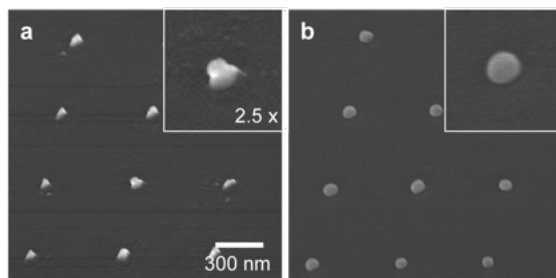


Figure S2. As-fabricated and thermal annealed patterned AuNPs. 45° tilted-view SEM images of (a) as-fabricated and (b) thermally annealed AuNPs. The tilted-view images show that as-fabricated AuNPs were nonuniform and had irregular cylindrical shapes. After annealing, the annealed AuNPs were uniform and had hemispherical shapes with smooth surfaces.

Morphology analysis of spiky AuNPs synthesized with different EPPS concentrations

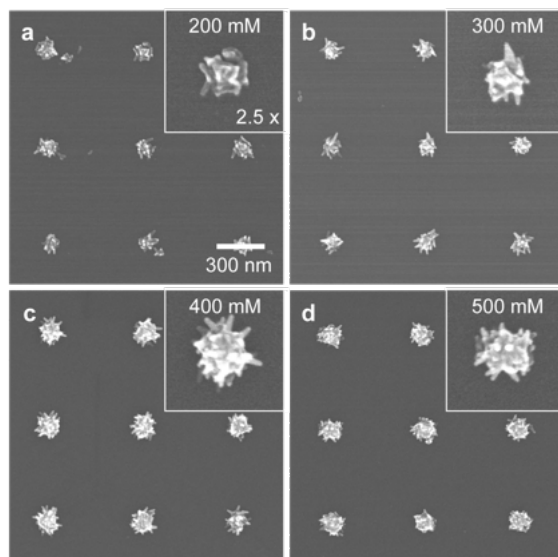


Figure S3. Spike growth on the patterned AuNPs with different EPPS concentrations. SEM images of spiky AuNPs synthesized with EPPS concentration of (a) 200 mM, (b) 300 mM, (c) 400 mM, and (d) 500 mM. The concentration of HAuCl₄ (0.4 mM) and solution pH (7.5) were kept consistent in different growth solutions.

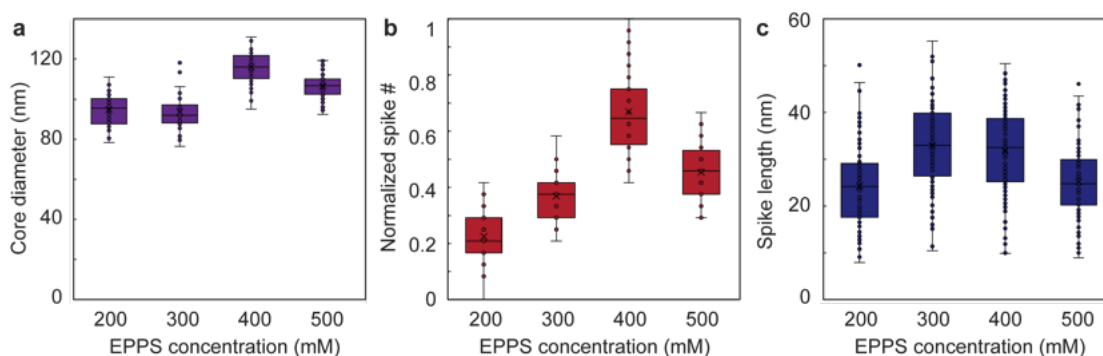


Figure S4. Morphology analysis of spike growth on the patterned AuNPs with different EPPS concentrations. (a) Core diameter, (b) normalized peripheral spike number, and (c) estimated spike length of the spiky AuNPs synthesized with EPPS concentrations of 200, 300, 400, and 500 mM. The concentration of HAuCl₄ (0.4 mM) and solution pH (7.5) were kept constant in different growth solutions.

Optical properties of spiky AuNPs synthesized with different EPPS concentrations

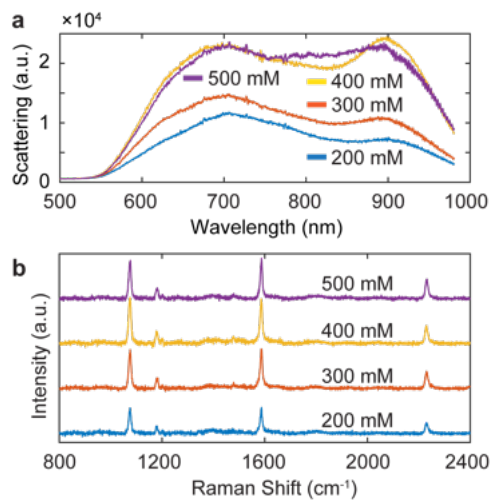


Figure S5. Optical properties of the patterned AuNPs after spike formation with different EPPS concentrations. (a) Dark-field scattering spectra of the patterned spiky AuNPs and (b) representative SERS spectra of 4-MBN monolayer on AuNPs synthesized with different EPPS concentrations (200, 300, 400 and 500 mM). The concentration of HAuCl₄ (0.4 mM) and solution pH (7.5) were kept constant in different growth solutions. The SERS spectra were measured with a 633-nm excitation laser.

Near-field enhancement of a single spiky AuNP at the LSP wavelengths

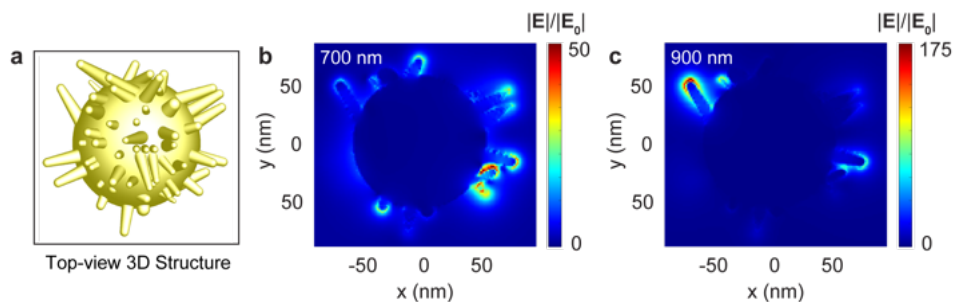


Figure S6. Electromagnetic near-fields of a single spiky AuNP at 700 and 900 nm. (a) Top-view 3D structure for FDTD simulation. Spatial distribution of electromagnetic near-field ($|E|/|E_0|$) of a single spiky AuNP based on the morphology of AuNPs synthesized with HAuCl_4 (0.4 mM) and EPPS (400 mM) at pH 7.5, using an excitation source with wavelengths of (b) 700 and (c) 900 nm. The x-y plane is 10 nm away from the quartz substrate surface. The excitation wavelengths of 700 nm was at the wavelength of the first localized surface plasmon (LSP) resonance, and 900 nm was at the wavelength of the second LSP resonance of the spiky AuNPs. The highest calculated electromagnetic field ($|E|/|E_0|$) is around 175 at the resonance wavelength of 900 nm. The highest SERS enhancement factor that a single spiky AuNP with the initial seed diameter of 70 nm is 9.5×10^8 at the resonance wavelength of 900 nm.

Morphology analysis of spiky AuNPs synthesized with different seed sizes

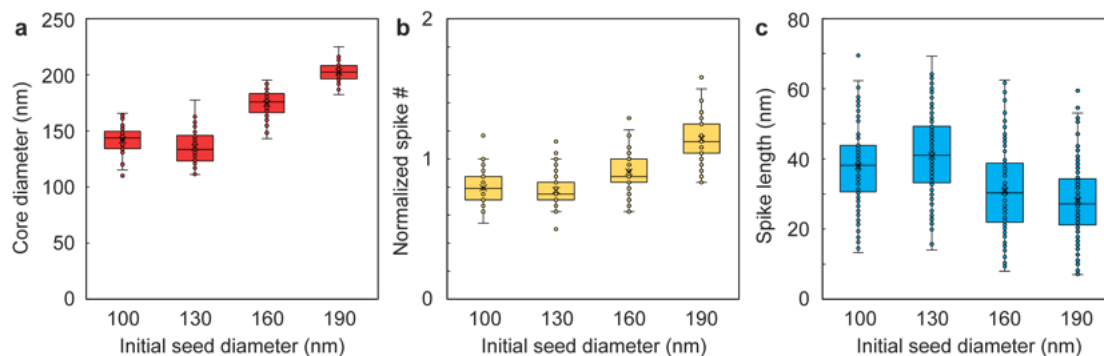


Figure S7. Morphology analysis of spike growth on the patterned AuNPs with different initial seed diameters. (a) Core diameter, (b) normalized peripheral spike number, and (c) estimated spike length of the patterned spiky AuNPs. The concentration of H₂AuCl₄ (0.4 mM) and EPPS (400 mM), and solution pH (7.5) were kept constant in different growth conditions.

Large-area spiky AuNP arrays

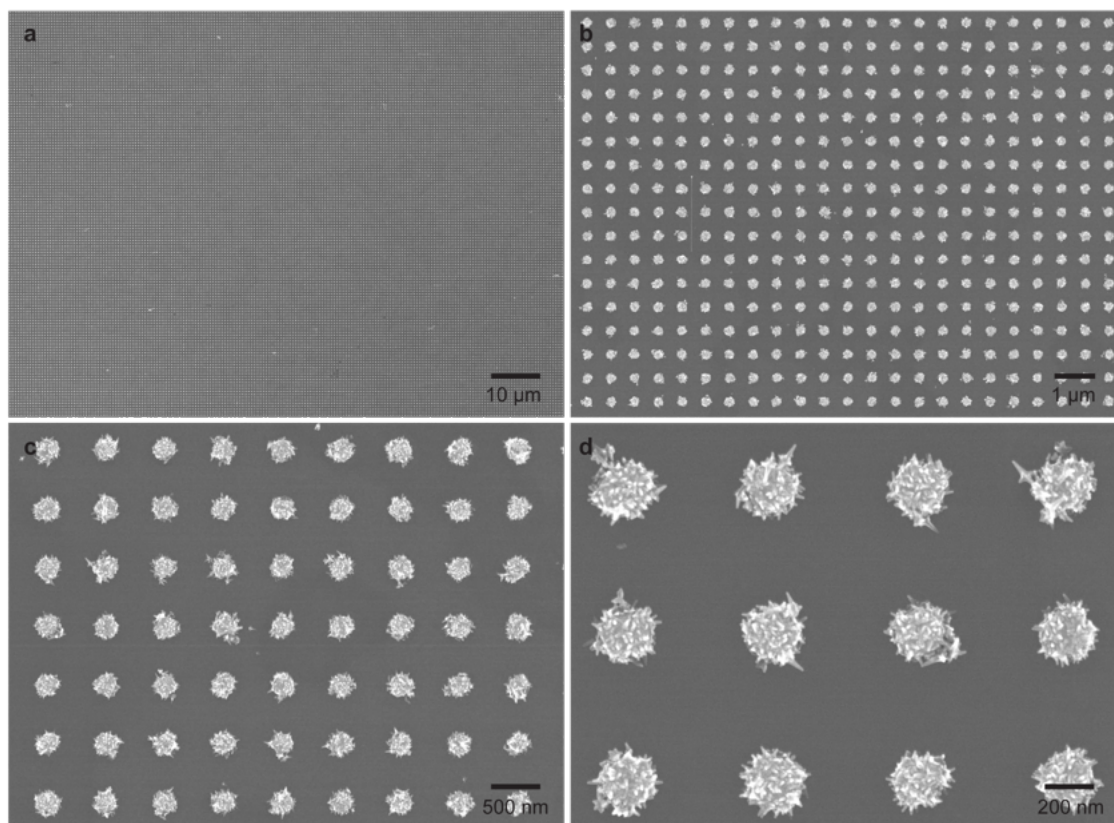


Figure S8. SEM images of the patterned spiky AuNPs with different magnifications: (a) $\times 1000$, (b) $\times 8000$, (c) $\times 20000$, and (d) $\times 50000$. The patterned AuNPs with diameters of 190 nm were used as the initial seeds.

Supporting Information references:

1. Choo, P.; Arenas-Esteban, D.; Jung, I.; Chang, W. J.; Weiss, E. A.; Bals, S.; Odom, T. W. Investigating Reaction Intermediates during the Seedless Growth of Gold Nanostars Using Electron Tomography. *ACS Nano* **2022**, *16* (3), 4408-4414. DOI: 10.1021/acsnano.1c10669.
2. Chandra, K.; Rugg, B. K.; Ratner, M. A.; Wasielewski, M. R.; Odom, T. W. Detecting and Visualizing Reaction Intermediates of Anisotropic Nanoparticle Growth. *J. Am. Chem. Soc.* **2018**, *140* (9), 3219-3222. DOI: 10.1021/jacs.8b00124.
3. Potter, R. F. Chapter 2 - Basic Parameters for Measuring Optical Properties. In *Handbook of Optical Constants of Solids*, Palik, E. D. Ed.; Academic Press, 1985; pp 11-34.
4. Lee, M. H.; Huntington, M. D.; Zhou, W.; Yang, J.-C.; Odom, T. W. Programmable Soft Lithography: Solvent-Assisted Nanoscale Embossing. *Nano Lett.* **2011**, *11* (2), 311-315. DOI: 10.1021/nl102206x.
5. Henzie, J.; Lee, M. H.; Odom, T. W. Multiscale Patterning of Plasmonic Metamaterials. *Nat. Nanotechn.* **2007**, *2* (9), 549-554. DOI: 10.1038/nnano.2007.252.